Infosys - Coding 16

HackerRank

You are given a permutation P and an integer K.

We define the number of ranges R for any array B as the minimum number of ranges that satisfy the following conditions:

1. Each element in B (by value) is included in exactly one range.
2. For each range [j, k] and each i such that (j ≤ i ≤ k), i appears as an element in B.

You need to choose exactly K non-intersecting subarrays in P. Let the sum of the value R for all the chosen subarrays be defined as X.

Find the maximum possible value of X in P.

Note: A permutation P is defined as a sequence of integers where (0 ≤ P[i] ≤ (N−1)) such that each element appears exactly once in P.

Input Format

The first line contains an integer, N, denoting the size of the permutation P.

The next line contains an integer, K, described in the problem.

Each line i of the N subsequent lines (where 0 ≤ i ≤ N−1) contains an integer describing P[i]. (this line was partly cut off at the bottom of image 1, inferred from the sample layout)

Constraints

1 <= K <= N

1 <= P[i] <= N

Output Format

The output contains an integer value

Sample Input 0

```
3
2
1
3
2
```

Sample Output 0

```
3
```

Explanation 0

N=3, K=2, P=[1,3,2] The optimal way is to choose subarrays [1,3] and [2]. The ranges in subarray [1,3] are [1,1] and [3,3]. The range in subarray [2] is [2,2]. The number of ranges: 2 + 1 = 3.

Sample Input 1

```
3
1
1
3
2
```

Sample Output 1

```
2
```

Explanation 1

N=3, K=1, P=[1,3,2] The optimal way is to choose the subarray [1,3]. The ranges representing it i.e. [1,1] and [3,3]. Hence the answer is 2.

Sample Input 2

```
4
3
1
4
2
3
```

Sample Output 2

```
4
```

Explanation 2

N=4, K=3, P=[1,4,2,3] The optimal way is to choose the 3 subarrays: [1], [4,2] and [3]. Subarray [1] can be represented by the range [1,1]. Subarray [4,2] can be represented by the ranges [4,4] and [2,2]. Subarray [3] can be represented by the range [3,3]. The number of ranges are 1 + 1 + 1 + 1 = 4.